

Q To find Shortest Common Supersequence

Soln

Step 1: Find LCS (Longest Common Subsequence)
Step 2: Use LCS to build Shortest Common Supersequence

$$SCS \text{ length} = n + m - LCS$$

Approach:

SCS = Shortest Common Supersequence

Build LCS DP Table

Let $dp[i][j]$ = LCS Length of $Str1[0 \dots i-1]$ and $Str2[0 \dots j-1]$

If Characters match:

$$dp[i][j] = 1 + dp[i-1][j-1]$$

Else

$$dp[i][j] = \max(dp[i-1][j], dp[i][j-1])$$

→ Back track to build SCS

Start from $(i=n, j=m)$

Case

① If Characters match

Add first Character only

move diagonally ($i--$, $j--$)

② if ^{not match}
if $dp[i-1][i] > dp[i][j-1]$

→ take $str 1[i-1]$, move $j--$

else

take $str 2[i-1]$, move $j--$

③ if one string finishes

• Append remaining characters

④ Return to result